

PRACTICAL - 5

Implementation of 0/1 Knapsack using Dynamic Programming

Aim of the Practical:

To solve the 0/1 Knapsack problem using dynamic programming and analyze the DP table construction.

Algorithm:

1. Let $dp[i][w]$ denote the best value using the first i items and capacity w .
2. Initialize row 0 and column 0 with 0.
3. If $wt[i-1] > w$, set $dp[i][w] = dp[i-1][w]$.
4. Otherwise set $dp[i][w] = \max(dp[i-1][w], value[i-1] + dp[i-1][w-wt[i-1]])$.
5. Return $dp[n][W]$.

Program:

```
#include <iostream>
#include <vector>
#include <algorithm>
using namespace std;
int knapsack(int W, const vector<int>& wt, const vector<int>& val) {
    int n = wt.size();
    vector<vector<int>>> dp(n + 1, vector<int>(W + 1, 0));
    for (int i = 1; i <= n; i++) {
        for (int w = 0; w <= W; w++) {
            dp[i][w] = dp[i - 1][w];
            if (wt[i - 1] <= w) {
                dp[i][w] = max(dp[i][w],
                    val[i - 1] + dp[i - 1][w - wt[i - 1]]);
            }
        }
    }
    cout << "DP Table:" << endl;
    for (int i = 0; i <= n; i++) {
        for (int w = 0; w <= W; w++)
            cout << dp[i][w] << " ";
        cout << endl;
    }
    return dp[n][W];
}
int main() {
    vector<int> wt = {1, 3, 4, 5};
    vector<int> val = {1, 4, 5, 7};
    int W = 7;
    cout << knapsack(W, wt, val);
}
```

Output:

```
DP Table:
0 0 0 0 0 0 0 0
0 1 1 1 1 1 1 1
0 1 1 4 5 5 5 5
0 1 1 4 5 6 6 9
0 1 1 4 5 7 8 9
9
```



Time Analysis:

Derivation of Time Complexity:

Let n be the number of items and W be the knapsack capacity.

The DP table contains $(n + 1)$ rows and $(W + 1)$ columns.

Number of states = $(n + 1)(W + 1)$

The two nested loops visit every non-base state once.

For each state, only constant-time comparisons, additions and max operations are performed.

$$T(n, W) = c \cdot n \cdot (W + 1) + c_0$$

$$T(n, W) = \Theta(nW)$$

The table stores $(n + 1)(W + 1)$ integer values.

$$S(n, W) = \Theta(nW)$$

This is pseudo-polynomial because W is a numeric capacity, not the number of bits needed to encode W .

DP Table Analysis:

Row 0 represents the base case with no items, so every value is 0.

Each next row shows the best value after considering one more item for capacities 0 to 7.

The final cell $dp[4][7] = 9$, which is the maximum value for capacity 7.

Final Time Analysis:

- Time Complexity: $\Theta(nW)$.
- Space Complexity: $\Theta(nW)$.

Result:

The 0/1 Knapsack problem was solved using dynamic programming, and its complexity was derived from the number of DP states.